

Confirmation

Along the way I saw enough of the confirmation error to make Karl Popper stand up with rage. People would find data in which there were no jumps or extreme events, and show me a “proof” that one could use the Gaussian. This was exactly like my example of the “proof” that O. J. Simpson is not a killer in [Chapter 5](#). The entire statistical business confused absence of proof with proof of absence. Furthermore, people did not understand the elementary asymmetry involved: you need one single observation to reject the Gaussian, but millions of observations will not fully confirm the validity of its application. Why? Because the Gaussian bell curve disallows large deviations, but tools of Extremistan, the alternative, do not disallow long quiet stretches.

I did not know that Mandelbrot’s work mattered outside aesthetics and geometry. Unlike him, I was not ostracized: I got a lot of approval from practitioners and decision makers, though not from their research staffs.

But suddenly I got the most unexpected vindication.

IT WAS JUST A BLACK SWAN

Robert Merton, Jr., and Myron Scholes were founding partners in the large speculative trading firm called Long-Term Capital Management, or LTCM, which I mentioned in [Chapter 4](#). It was a collection of people with top-notch résumés, from the highest ranks of academia. They were considered geniuses. The ideas of portfolio theory inspired their risk management of possible outcomes—thanks to their sophisticated “calculations.” They managed to enlarge the ludic fallacy to industrial proportions.

Then, during the summer of 1998, a combination of large events, triggered by a Russian financial crisis, took place that lay outside their models. It was a Black Swan. LTCM went bust and almost took down the entire financial system with it, as the exposures were massive. Since their models ruled out the possibility of large deviations, they allowed themselves to take a monstrous amount of risk. The ideas of Merton and Scholes, as well as those of Modern Portfolio Theory, were starting to go bust. The magnitude of the losses was spectacular, too spectacular to allow us to ignore the intellectual comedy. Many friends and I thought that the portfolio theorists would suffer the fate of tobacco companies: they were endangering people’s savings and would soon be brought to account for the consequences of their Gaussian-inspired methods.

None of that happened.

Instead, MBAs in business schools went on learning portfolio theory. And